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Conveyance device

Abstract

A conveyance device includes a magnetic scale including linear movement magnetic scales formed linearly along a linear direction, and curvilinear movement magnetic scales formed curvilinearly along a curvilinear direction. The linear movement magnetic scales and the curvilinear movement magnetic scales are spaced apart from each other.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a National Stage of International Patent Application No.

PCT/JP2021/031761, filed Aug. 30, 2021, the entire content of which is incorporated herein by reference.

BACKGROUND

Technical Field

(2) The present disclosure relates to a conveyance device, and more particularly, it relates to a conveyance device including a slider and a drive module.

Background Art

(3) Conventionally, a conveyance device including a slider and a drive module is known. Such a conveyance device is disclosed in International Publication No. 2019/007199, for example.

(4) International Publication No. 2019/007199 discloses a linear conveyance device including a movable body (slider) and a stator module (drive module). The movable body moves linearly or curvilinearly due to a magnetic field generated in the stator module.

(5) The movable body disclosed in International Publication No. 2019/007199 includes a base, a first magnetic generator, and a second magnetic generator. The first magnetic generator and the second magnetic generator are attached to the base while being adjacent to each other and in contact with each other. The first magnetic generator generates magnetism to detect the position of the movable body during linear movement. The second magnetic generator generates magnetism to detect the position of the movable body during curvilinear movement.

(6) The stator module disclosed in International Publication No. 2019/007199 includes a magnetic reader that reads the magnetism of each of the first magnetic generator and the second magnetic generator. In the linear conveyance device, the position of the movable body is detected based on the magnetism of each of the first magnetic generator and the second magnetic generator read by the magnetic reader.

SUMMARY

(7) In the linear conveyance device disclosed in International Publication No. 2019/007199, the first magnetic generator and the second magnetic generator are attached to the base while being adjacent to each other and in contact with each other. Therefore, in the linear conveyance device disclosed in International Publication No. 2019/007199, magnetic interference is conceivably likely to occur between the first magnetic generator and the second magnetic generator. Thus, in the linear conveyance device disclosed in International Publication No. 2019/007199, it is desired to more reliably reduce or prevent magnetic interference between the first magnetic generator (linear movement magnetic scale) and the second magnetic generator (curvilinear movement magnetic scale).

(8) Accordingly, the present disclosure provides a conveyance device capable of more reliably reducing or preventing magnetic interference between a linear movement magnetic scale and a curvilinear movement magnetic scale.

(9) A conveyance device according to an aspect of the present disclosure includes a slider movable along a linear direction and a curvilinear direction, a drive module to move the slider along the linear direction or the curvilinear direction, a magnetic scale including linear movement magnetic scales formed linearly along the linear direction, and curvilinear movement magnetic scales formed curvilinearly along the curvilinear direction and provided on either the slider or the drive module, and magnetic sensors provided in either the drive module or the slider to detect magnetism of the linear movement magnetic scales and the curvilinear movement magnetic scales, respectively. The linear movement magnetic scales and the curvilinear movement magnetic scales are spaced apart from each other.

(10) In the conveyance device according to this aspect of the present disclosure, as described above, the linear movement magnetic scales and the curvilinear movement magnetic scales are

spaced apart from each other. Accordingly, a distance between the linear movement magnetic scales and the curvilinear movement magnetic scales can be ensured, and thus magnetic interference between the linear movement magnetic scales and the curvilinear movement magnetic scales can be more reliably reduced or prevented.

(11) In the conveyance device according to this aspect, the linear movement magnetic scales and the curvilinear movement magnetic scales are preferably provided on a same surface or different surfaces and are preferably spaced apart from each other. Accordingly, the linear movement magnetic scales and the curvilinear movement magnetic scales can be provided on the same surface or the different surfaces of the same member, and thus as compared with a case in which the linear movement magnetic scales and the curvilinear movement magnetic scales are provided on surfaces of different members, an increase in the number of components of the conveyance device can be reduced or prevented.

(12) In the conveyance device according to this aspect, the magnetic scale is preferably provided on the slider, and the magnetic sensors are preferably provided in the drive module. Accordingly, the drive module is fixed at a predetermined location, and thus unlike a case in which the magnetic sensors are provided in a movable structure like the slider, power supply equipment for supplying power to the magnetic sensors can be easily connected to the magnetic sensors.

(13) In this case, the slider preferably includes a slider frame in which the magnetic scale is located such that the linear movement magnetic scales and the curvilinear movement magnetic scales face the magnetic sensors in an upward-downward direction or in a width direction orthogonal to the upward-downward direction and a moving direction of the slider. Accordingly, the magnetic scale can be housed inside the slider frame, and thus as compared with a case in which the magnetic scale is located outside the slider frame having the same size, an increase in the size of the slider can be reduced or prevented.

(14) In the conveyance device according to this aspect, the magnetic scale preferably further includes a base with which the linear movement magnetic scales and the curvilinear movement magnetic scales are provided integrally. Accordingly, the linear movement magnetic scales and the curvilinear movement magnetic scales are provided integrally with the same base, and thus even when the base expands or contracts due to temperature changes, errors can be caused in measurement of the position of the slider according to the expansion and contraction of the base with respect to the linear movement magnetic scales and the curvilinear movement magnetic scales. Therefore, the errors in position measurement of the linear movement magnetic scales and the curvilinear movement magnetic scales can be substantially equal to each other. Consequently, the same correction value can be used to correct errors in position measurement that occur when the slider moves linearly and errors in position measurement that occur when the slider moves curvilinearly, and thus a control for position acquisition can be facilitated.

(15) In this case, the base includes a first surface on which the linear movement magnetic scales are provided, and a second surface that is different from the first surface and on which the curvilinear movement magnetic scales are provided. Accordingly, as compared with a case in which the linear movement magnetic scales and the curvilinear movement magnetic scales are aligned on the same surface, an increase in the dimension of the base in a direction in which the linear movement magnetic scales and the curvilinear movement magnetic scales are aligned can be reduced or prevented, and thus an increase in the size of the conveyance device in the alignment direction described above can be reduced or prevented.

(16) In the conveyance device including the second surface different from the first surface, the second surface is preferably arranged on a back side of the first surface in the base. Accordingly, the first surface constitutes the front surface of the base, and the second surface constitutes the back surface of the base such that in a plan view, the linear movement magnetic scales and the curvilinear movement magnetic scales can overlap each other. Thus, an increase in the dimension of the base in the moving direction of the slider can be reduced or prevented.

(17) In the conveyance device in which the second surface is arranged on the back side of the first surface in the base, the magnetic scale preferably includes a magnetic shielding plate located between the linear movement magnetic scales and the curvilinear movement magnetic scales to block magnetism emitted from each of the linear movement magnetic scales and the curvilinear movement magnetic scales. Accordingly, magnetic interference between the linear movement magnetic scales and the curvilinear movement magnetic scales can be reduced or prevented by the magnetic shielding plate, and thus a distance by which the linear movement magnetic scales and the curvilinear movement magnetic scales are spaced apart from each other can be reduced. Consequently, thicknesses required for the bases arranged between the linear movement magnetic scales and the curvilinear movement magnetic scales can be reduced, and thus an increase in the size of the magnetic scale including the linear movement magnetic scales, the curvilinear movement magnetic scales, and the bases can be reduced or prevented.

(18) In the conveyance device including the first surface and the second surface, the second surface and the first surface are preferably substantially orthogonal to each other. Accordingly, the linear movement magnetic scales are provided on the first surface, and the curvilinear movement magnetic scales are provided on the second surface that is substantially orthogonal to the first surface, and thus the magnetic sensor that detects the magnetism of the linear movement magnetic scales can be placed in the vicinity of the magnetic sensor that detects the magnetism of the curvilinear movement magnetic scales. Consequently, a space required to place the magnetic sensor that detects the magnetism of the linear movement magnetic scales and the magnetic sensor that detects the magnetism of the curvilinear movement magnetic scales can be made compact.

(19) In the conveyance device in which the second surface is orthogonal to the first surface, the second surface and the first surface are preferably substantially orthogonal to each other, and the second surface is preferably a curved surface corresponding to a curvature of the curvilinear movement magnetic scales. Accordingly, the curved surface of the second surface is formed along the curvilinear direction of the slider, and thus the position of the slider during curvilinear movement can be appropriately detected by the curvilinear movement magnetic scales provided on the second surface.

(20) In the conveyance device including the base, the linear movement magnetic scales and the curvilinear movement magnetic scales are preferably provided on a same surface of the base and are preferably spaced apart from each other in a direction in which the linear movement magnetic scales and the curvilinear movement magnetic scales are aligned. Accordingly, as compared with a case in which the linear movement magnetic scales and the curvilinear movement magnetic scales are aligned in a direction perpendicular to the same surface, an increase in a space for installing the linear movement magnetic scales and the curvilinear movement magnetic scales in the direction perpendicular to the same surface can be reduced or prevented.

(21) In the conveyance device according to this aspect, the magnetic scale preferably further includes a base member in which the linear movement magnetic scales and the curvilinear movement magnetic scales are provided, the curvilinear movement magnetic scales preferably include first curvilinear scales formed along a first curvilinear direction as the curvilinear direction, and second curvilinear scales formed along a second curvilinear direction different from the first curvilinear direction, and the base member preferably includes a linear position detection surface that constitutes an upper surface or a lower surface of the base member and on which the linear movement magnetic scales are provided, a first curvilinear position detection surface that protrudes from the linear position detection surface and on which the first curvilinear scales are provided, and a second curvilinear position detection surface that faces the first curvilinear position detection surface and on which the second curvilinear scales are provided. Accordingly, with the first curvilinear scales and the second curvilinear scales, the position of the slider can be measured when the slider moves curvilinearly in a plurality of curvilinear directions including the first curvilinear direction and the second curvilinear direction, and thus the position measurement can be performed

even when the slider moves curvilinearly in curvilinear directions that are opposite to each other.

Thus, the degree of freedom of a path along which the slider moves can be improved.

(22) In the conveyance device according to this aspect, the magnetic scale is preferably provided on the slider, and in a moving direction of the slider, first ends of the linear movement magnetic scales and first ends of the curvilinear movement magnetic scales are preferably located at substantially a same position, and second ends of the linear movement magnetic scales and second ends of the curvilinear movement magnetic scales are preferably located at substantially a same position.

Accordingly, the lengths of the linear movement magnetic scales and the curvilinear movement magnetic scales can be substantially equal to each other in the moving direction of the slider, and thus when either the linear movement magnetic scales or the curvilinear movement magnetic scales are set to the maximum length according to the installation space, either the curvilinear movement magnetic scales or the linear movement magnetic scales can have the same maximum length.

Consequently, the magnetic sensors are arranged such that a distance between the magnetic sensors matches the maximum length, and thus the magnetic sensors can be aligned with a maximum distance between the magnetic sensors. Thus, the number of magnetic sensors required to measure the position of the slider can be minimized.

(23) According to the present disclosure, as described above, it is possible to more reliably reduce or prevent magnetic interference between the linear movement magnetic scales and the curvilinear movement magnetic scales.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a plan view showing a conveyance device according to a first embodiment.

(2) FIG. 2 is a sectional view taken along the line **111-111** in FIG. 1;

(3) FIG. 3 is a sectional view taken along the line **112-112** in FIG. 1;

(4) FIG. 4 is a top view of a magnetic scale of the conveyance device according to the first embodiment;

(5) FIG. 5 is a bottom view of the magnetic scale of the conveyance device according to the first embodiment;

(6) FIG. 6 is an enlarged view of a portion D in FIG. 3;

(7) FIG. 7 is a schematic view showing a state in which a slider of the conveyance device according to the first embodiment is moving on a linear portion;

(8) FIG. 8 is a schematic view showing a state in which the slider of the conveyance device according to the first embodiment is moving on a curvilinear portion;

(9) FIG. 9 is an enlarged view of a portion of a conveyance device according to a second embodiment, corresponding to the portion D in FIG. 3;

(10) FIG. 10 is a top view of a magnetic scale of a conveyance device according to a third embodiment;

(11) FIG. 11 is a perspective view showing a magnetic scale of a conveyance device according to a fourth embodiment;

(12) FIG. 12 is a perspective view showing a magnetic scale of a conveyance device according to a fifth embodiment;

(13) FIG. 13 is a perspective view showing a back yoke and a magnetic scale of a conveyance device according to a first modified example of the first embodiment;

(14) FIG. 14 is a sectional view showing a magnetic scale and a magnetic shielding plate of a conveyance device according to a second modified example of the second embodiment;

(15) FIG. 15 is a sectional view showing a magnetic scale and a magnetic shielding plate of a conveyance device according to a third modified example of the second embodiment; and

(16) FIG. 16 is a perspective view showing the magnetic scale and the magnetic shielding plate of the conveyance device according to the third modified example of the second embodiment.

DETAILED DESCRIPTION

(17) Embodiments embodying the present disclosure are hereinafter described on the basis of the drawings.

First Embodiment

(18) The structure of a conveyance device **100** according to a first embodiment of the present disclosure is described with reference to FIGS. 1 to 8.

(19) Structure of Conveyance Device

(20) As shown in FIGS. 1 and 2, the conveyance device **100** is installed on the upper surface **101a** of a platform **101**. Although not shown, various robots are installed on the platform **101** to carry out tasks such as transferring workpieces such as boards and mounting electronic components or the like onto the boards. The various robots are arranged along a circular path **102** for sliders **3** described below. The conveyance device **100** is a linear conveyor that moves the sliders **3** along the circular path **102** using a linear motor.

(21) The conveyance device **100** moves the sliders **3** toward the work positions of the various robots. Specifically, the conveyance device **100** includes a plurality of linear movement drive modules **1**, a plurality of curvilinear movement drive modules **2**, a plurality of sliders **3**, magnetic scales **4**, magnetic sensors **5**, and magnetic sensors **6**. The plurality of linear movement drive modules **1** and the plurality of curvilinear movement drive modules **2** form the circular path **102** for the sliders **3**. The linear movement drive modules **1** and the curvilinear movement drive modules **2** are examples of a “drive module” in the claims.

(22) In horizontal directions, a direction in which the linear movement drive modules **1** extend is defined as an X direction, one direction in the X direction is defined as an X1 direction, and the other direction in the X direction is defined as an X2 direction. In the horizontal directions, a direction orthogonal to the X direction is defined as a Y direction, one direction in the Y direction is defined as a Y1 direction, and the other direction in the Y direction is defined as a Y2 direction. A direction orthogonal to the X direction and the Y direction is defined as a Z direction (upward-downward direction), one direction in the Z direction is defined as a Z1 direction (upward direction), and the other direction in the Z direction is defined as a Z2 direction (downward direction). Furthermore, the moving direction of the sliders **3** is defined as an M direction, one direction in the M direction is defined as an M1 direction, and the other direction in the M direction is defined as an M2 direction. A width direction orthogonal to the M direction and the Z direction is defined as a W direction, one direction in the W direction is defined as a W1 direction, and the other direction in the W direction is defined as a W2 direction.

(23) Each of the plurality of linear movement drive modules **1** is a linear conveyor module. The plurality of linear movement drive modules **1** are arranged on the platform **101**. The plurality of linear movement drive modules **1** are linearly connected to each other in the X direction. The plurality of linear movement drive modules **1** are arranged on the Y1 direction side and the Y2 direction side. The plurality of linear movement drive modules **1** have the same structure, and thus one linear movement drive module **1** arranged on the Y2 direction side and the X1 direction side is described.

(24) The linear movement drive module **1** is a device that moves the sliders **3** along a linear direction in the X1 direction or the X2 direction. The linear movement drive module **1** stops the sliders **3** at the work positions of the robots and moves the sliders **3** toward the work positions of the next robots after the work. Furthermore, the linear movement drive module **1** can move the sliders **3** while performing a position/speed control during the work, while also moving the sliders **3** toward the work positions of the next robots after the work.

(25) Specifically, as shown in FIG. 2, the linear movement drive module **1** includes a support member **11**, a guide rail **12**, and a linear motor stator **13**.

(26) The support member **11** supports the guide rail **12** and the linear motor stator **13**. The support member **11** is made of metal such as aluminum. Thus, the support member **11** is a metal frame. Although an example has been shown in which the support member **11** according to this embodiment is made of metal, the support member **11** may be made of resin.

(27) The guide rail **12** has a function of guiding the sliders **3**. The guide rail **12** is arranged on a central portion of the upper surface of the support member **11** in the Y direction. The guide rail **12** extends in the X direction. The linear motor stator **13** includes a plurality of electromagnets. Although an example has been shown in which the guide rail **12** according to this embodiment is arranged on the central portion of the upper surface of the support member **11** in the Y direction, the guide rail **12** may be arranged on a position offset from the central portion of the upper surface of the support member **11** in the Y direction. Furthermore, the guide rail **12** may be provided on a side surface of the support member **11** instead of on the upper surface of the support member **11**.

(28) As shown in FIG. 1, each of the plurality of curvilinear movement drive modules **2** is a linear conveyor module. The plurality of curvilinear movement drive modules **2** are arranged on the platform **101**. The plurality of curvilinear movement drive modules **2** are connected to each other along a curvilinear direction. The plurality of curvilinear movement drive modules **2** are arranged on the X1 direction side and the X2 direction side on the platform **101**. The plurality of curvilinear movement drive modules **2** have the same structure, and thus one curvilinear movement drive module **2** arranged on the X1 direction side is described. The curvilinear direction refers to a direction along an R direction (circumferential direction) around a central axis C1 (or C2) extending in the Z direction.

(29) The curvilinear movement drive module **2** is a device that moves the sliders **3** along the R1 direction or the R2 direction (curvilinear direction). The curvilinear movement drive module **2** stops the sliders **3** at the work positions of the robots and moves the sliders **3** toward the work positions of the next robots after the work. Furthermore, the curvilinear movement drive module **2** can move the sliders **3** while performing a position/speed control during the work, while also moving the sliders **3** toward the work positions of the next robots after the work.

(30) Specifically, as shown in FIG. 3, the curvilinear movement drive module **2** includes a support member **21**, a guide rail **22**, and a linear motor stator **23**.

(31) The support member **21** supports the guide rail **22** and the linear motor stator **23**. The support member **21** is made of metal such as aluminum. Thus, the support member **21** is a metal frame. Although an example has been shown in which the support member **21** according to this embodiment is made of metal, the support member **21** may be made of resin.

(32) The guide rail **22** has a function of guiding the sliders **3**. The guide rail **22** is arranged on a central portion of the upper surface of the support member **21** in a direction orthogonal to the central axis C1. The guide rail **22** extends along the R direction. The linear motor stator **23** includes a plurality of electromagnets. Although an example has been shown in which the guide rail **22** according to this embodiment is arranged on the central portion of the upper surface of the support member **21** in the direction orthogonal to the central axis C1, the guide rail **22** may be arranged on a position offset from the central portion of the upper surface of the support member **21** in the direction orthogonal to the central axis C1. Furthermore, the guide rail **22** may be provided on a side surface of the support member **21** instead of on the upper surface of the support member **21**.

(33) As shown in FIG. 1, the plurality of sliders **3** are arranged on the circular path **102**. The plurality of sliders **3** all have the same structure, and thus one of the plurality of sliders **3** is described.

(34) The slider **3** is movable along the linear direction and the curvilinear direction. The slider **3** moves in the linear direction by being guided by the guide rail **12**. The slider **3** moves in the curvilinear direction by being guided by the guide rail **22**. The linear direction refers to a direction of movement along the X direction. The curvilinear direction refers to a direction of movement along the R direction.

(35) Thus, the slider **3** slides linearly and curvilinearly on the circular path **102**. Specifically, as shown in FIGS. **2** and **3**, the slider **3** includes a slider frame **31**, a linear motor mover **32**, and a bracket **33**.

(36) The slider frame **31** is made of a metal material such as aluminum. The linear motor mover **32** is provided on the W1 direction side on an inner upper surface **36** provided between a pair of inner side surfaces **34** and **35** of the slider frame **31** facing each other in the W direction. The linear motor mover **32** includes a back yoke (not shown) and permanent magnets (not shown). The back yoke is a member that holds the permanent magnets and forms a magnetic path, and is made of a magnetic iron-based material such as iron or steel. The permanent magnets are arranged in the M direction such that their north and south poles are alternately directed in the Z2 direction. The permanent magnets face the linear motor stator **13** in the Z direction. The bracket **33** is a member for fixing the magnetic scale **4** to the slider frame **31**. The bracket **33** is provided on the inner surface **34** on the W2 direction side among the pair of inner side surfaces **34** and **35**. The bracket **33** is fixed to the slider frame **31** by a fastening member.

(37) In the slider **3**, a propulsive force is generated by the magnetic interaction between the magnetic flux generated in a coil when currents with different phases are supplied to the linear motor stator **13** (**23**) and the magnetic flux of the permanent magnets of the linear motor mover **32**. That is, the slider **3** is movable in the X direction or the R direction by the propulsive force described above.

(38) Magnetic Scale

(39) As shown in FIGS. **2** and **3**, the magnetic scale **4** is a scale for measuring the position of the slider **3** that slides linearly and curvilinearly. The magnetic scale **4** is provided on the slider **3**. The magnetic scale **4** is provided on the slider frame **31** via the bracket **33**. The magnetic scale **4** is arranged inside the slider frame **31** with linear movement magnetic scales **41** facing the magnetic sensors **5** in the Z direction, and curvilinear movement magnetic scales **42** facing the magnetic sensors **6** in the Z direction.

(40) The magnetic scale **4** includes a base **40**, the linear movement magnetic scales **41**, and the curvilinear movement magnetic scales **42**.

(41) Base

(42) The base **40** is a plate-shaped member made of a magnetic material using a ferrite material, for example. That is, the base **40** has a rectangular shape, the longitudinal direction of which is in the X direction as viewed from the Z1 direction side. The base **40** has a rectangular parallelepiped shape. The thickness of the base **40** in the Z direction is smaller than the width of the base **40** in the Y direction. The linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are recorded on the base **40** by being magnetized.

(43) Linear Movement Magnetic Scale and Curvilinear Movement Magnetic Scale

(44) As shown in FIGS. **2** and **4**, the linear movement magnetic scales **41** are recorded on the base **40** to measure the position of the slider **3** when the slider **3** is moving on the linear movement drive module **1**. The linear movement magnetic scales **41** are provided integrally with the base **40**.

Although the linear movement magnetic scales **41** are shown by broken lines in FIGS. **2** and **4**, this is merely an image for visualizing the linear movement magnetic scales **41** in the figures.

(45) The linear movement magnetic scales **41** are formed linearly along the linear direction (X direction). The linear movement magnetic scales **41** each have a length L1 in the X direction (the moving direction (M direction) of the slider **3**). A plurality of (three) linear movement magnetic scales **41** are provided on the base **40**. The plurality of linear movement magnetic scales **41** have different magnetization patterns. That is, in the plurality of linear movement magnetic scales **41**, the periods and phases of the magnetization patterns are different from each other. Two, or four or more linear movement magnetic scales **41** may be provided on the base **40** instead of three.

(46) The plurality of linear movement magnetic scales **41** are used to identify the plurality of sliders **3** and detect the positions of the plurality of sliders **3**. The positions of the sliders **3** are set in

correspondence with a difference between detection values of the magnetism of the plurality of linear movement magnetic scales **41**, the magnetization patterns of which have different periods and phases.

(47) As shown in FIGS. **3** and **5**, the curvilinear movement magnetic scales **42** are recorded on the base **40** to measure the position of the slider **3** when the slider **3** is moving on the curvilinear movement drive module **2**. The curvilinear movement magnetic scales **42** are provided integrally with the base **40**. Although the curvilinear movement magnetic scales **42** are shown by broken lines in FIGS. **3** and **5**, this is merely an image for visualizing the curvilinear movement magnetic scales **42** in the figures.

(48) The curvilinear movement magnetic scales **42** are formed curvilinearly along the curvilinear direction (R direction). The curvilinear movement magnetic scales **42** each have a length L2 in the M direction. A plurality of (three) curvilinear movement magnetic scales **42** are provided on the base **40**. The plurality of curvilinear movement magnetic scales **42** have different magnetization patterns. That is, in the plurality of curvilinear movement magnetic scales **42**, the periods and phases of the magnetization patterns are different from each other. Two, or four or more curvilinear movement magnetic scales **42** may be provided on the base **40** instead of three.

(49) The plurality of curvilinear movement magnetic scales **42** are used to identify the plurality of sliders **3** and detect the positions of the plurality of sliders **3**. The positions of the sliders **3** are set in correspondence with a difference between detection values of the magnetism of the plurality of curvilinear movement magnetic scales **42**, the magnetization patterns of which have different periods and phases.

(50) Positional Relationship Between Linear Movement Magnetic Scale and Curvilinear Movement Magnetic Scale

(51) In this embodiment, as shown in FIG. **6**, the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are provided integrally with the base **40** and are spaced apart from each other. The linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are provided integrally with the base **40** on different surfaces and are spaced apart from each other. Although the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are shown by broken lines in FIG. **6**, this is merely an image for visualizing the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** in the figure.

(52) Specifically, the linear movement magnetic scales **41** are provided integrally with the base **40** on a first surface **40a**. The first surface **40a** is the upper surface of the base **40** on the Z1 direction side. The linear movement magnetic scales **41** face the magnetic sensors **5** in the Z direction (see FIG. **2**). The curvilinear movement magnetic scales **42** are provided integrally with the base **40** on a second surface **40b**. The second surface **40b** is the lower surface of the base **40** on the Z2 direction side. Thus, the second surface **40b** is arranged on the back side of the first surface **40a** in the base **40**. The curvilinear movement magnetic scales **42** face the magnetic sensors **6** in the Z direction (see FIG. **3**).

(53) As shown in FIGS. **7** and **8**, in the moving direction (M direction) of the slider **3**, a first end **41a** of the linear movement magnetic scale **41** and a first end **42a** of the curvilinear movement magnetic scale **42** are located at substantially the same position. The first end **41a** of the linear movement magnetic scale **41** is an end of the linear movement magnetic scale **41** on the M1 direction side. The first end **42a** of the curvilinear movement magnetic scale **42** is an end of the curvilinear movement magnetic scale **42** on the M1 direction side.

(54) A second end **41b** of the linear movement magnetic scale **41** and a second end **42b** of the curvilinear movement magnetic scale **42** are located at substantially the same position. The second end **41b** of the linear movement magnetic scale **41** is an end of the linear movement magnetic scale **41** on the M2 direction side. The second end **42b** of the curvilinear movement magnetic scale **42** is an end of the curvilinear movement magnetic scale **42** on the M2 direction side.

(55) Magnetic Sensors

(56) As shown in FIG. 7, the magnetic sensors 5 are provided in the linear movement drive module 1. The magnetic sensors 5 detect the magnetism of the linear movement magnetic scales 41. A plurality of magnetic sensors 5 are arranged at a predetermined pitch in the M direction in accordance with the lengths L1 (see FIG. 4) of the linear movement magnetic scales 41 in the X direction. The predetermined pitch is an interval smaller than the length L1. The plurality of (three) magnetic sensors 5 are arranged at a predetermined pitch in the W direction so as to face the plurality of linear movement magnetic scales 41.

(57) As shown in FIG. 8, the magnetic sensors 6 are provided in the curvilinear movement drive module 2. The magnetic sensors 6 detect the magnetism of the curvilinear movement magnetic scales 42. A plurality of magnetic sensors 6 are arranged at a predetermined pitch in the R direction in accordance with the lengths L2 (see FIG. 5) of the curvilinear movement magnetic scales 42 in the M direction. The predetermined pitch is an interval smaller than the length L2. The plurality of (three) magnetic sensors 6 are arranged at a predetermined pitch in the direction orthogonal to the central axis C1 (C2) so as to face the plurality of curvilinear movement magnetic scales 42.

(58) According to the first embodiment, the following advantageous effects are achieved.

(59) According to the first embodiment, as described above, the linear movement magnetic scales 41 and the curvilinear movement magnetic scales 42 are spaced apart from each other. Accordingly, a distance between the linear movement magnetic scales 41 and the curvilinear movement magnetic scales 42 can be ensured, and thus magnetic interference between the linear movement magnetic scales 41 and the curvilinear movement magnetic scales 42 can be more reliably reduced or prevented.

(60) According to the first embodiment, as described above, the linear movement magnetic scales 41 and the curvilinear movement magnetic scales 42 are provided on the different surfaces and are spaced apart from each other. Accordingly, the linear movement magnetic scales 41 and the curvilinear movement magnetic scales 42 can be provided on the different surfaces of the same member, and thus as compared with a case in which the linear movement magnetic scales 41 and the curvilinear movement magnetic scales 42 are provided on surfaces of different members, an increase in the number of components of the conveyance device 100 can be reduced or prevented.

(61) According to the first embodiment, as described above, the magnetic scale 4 is provided on the slider 3. Moreover, the magnetic sensors 5 and the magnetic sensors 6 are provided in the linear movement drive module 1 and the curvilinear movement drive module 2, respectively.

Accordingly, the linear movement drive module 1 and the curvilinear movement drive module 2 are fixed at predetermined locations, and thus unlike a case in which the magnetic sensors 5 (8) are provided in a movable structure like the slider 3, power supply equipment for supplying power to the magnetic sensors 5 (8) can be easily connected to the magnetic sensors 5 (8).

(62) According to the first embodiment, as described above, the slider 3 includes the slider frame 31 in which the magnetic scale 4 is located such that the linear movement magnetic scales 41 and the curvilinear movement magnetic scales 42 face the magnetic sensors 5 and the magnetic sensors 8, respectively, in the upward-downward direction. Accordingly, the magnetic scale 4 can be housed inside the slider frame 31, and thus as compared with a case in which the magnetic scale 4 is located outside the slider frame 31 having the same size, an increase in the size of the slider 3 can be reduced or prevented.

(63) According to the first embodiment, as described above, the magnetic scale 4 includes the base 40 with which the linear movement magnetic scales 41 and the curvilinear movement magnetic scales 42 are provided integrally. Accordingly, the linear movement magnetic scales 41 and the curvilinear movement magnetic scales 42 are provided integrally with the same base 40, and thus even when the base 40 expands or contracts due to temperature changes, errors can be caused in measurement of the position of the slider 3 according to the expansion and contraction of the base 40 with respect to the linear movement magnetic scales 41 and the curvilinear movement magnetic

scales **42**. Therefore, the errors in position measurement of the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** can be substantially equal to each other. Consequently, the same correction value can be used to correct errors in position measurement that occur when the slider **3** moves linearly and errors in position measurement that occur when the slider **3** moves curvilinearly, and thus a control for position acquisition can be facilitated.

(64) According to the first embodiment, as described above, the base **40** includes the first surface **40a** on which the linear movement magnetic scales **41** are provided, and the second surface **40b** that is different from the first surface **40a** and on which the curvilinear movement magnetic scales **42** are provided. Accordingly, as compared with a case in which the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are aligned on the same surface, an increase in the dimension of the base **40** in a direction in which the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are aligned can be reduced or prevented, and thus an increase in the size of the conveyance device **100** in the alignment direction described above can be reduced or prevented.

(65) According to the first embodiment, as described above, the second surface **40b** is arranged on the back side of the first surface **40a** in the base **40**. Accordingly, the first surface **40a** constitutes the front surface of the base **40**, and the second surface **40b** constitutes the back surface of the base **40** such that in a plan view, the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** can overlap each other. Thus, an increase in the dimension of the base **40** in the M direction can be reduced or prevented.

(66) According to the first embodiment, as described above, in the moving direction of the slider **3**, the first end **41a** of the linear movement magnetic scale **41** and the first end **42a** of the curvilinear movement magnetic scale **42** are located at substantially the same position, and the second end **41b** of the linear movement magnetic scale **41** and the second end **42b** of the curvilinear movement magnetic scale **42** are located at substantially the same position. Accordingly, the lengths L1 and L2 of the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** can be substantially equal to each other in the M direction (the moving direction of the slider **3**), and thus when either the linear movement magnetic scales **41** or the curvilinear movement magnetic scales **42** are set to the maximum length according to the installation space, either the curvilinear movement magnetic scales **42** or the linear movement magnetic scales **41** can have the same maximum length. Consequently, the magnetic sensors **5 (8)** are arranged such that a distance between the magnetic sensors **5 (8)** matches the maximum length, and thus the magnetic sensors **5 (8)** can be aligned with a maximum distance between the magnetic sensors **5 (8)**. Thus, the number of magnetic sensors **5** and the number of magnetic sensors **6** required to measure the position of the slider **3** can be minimized.

Second Embodiment

(67) The structure of a magnetic scale **204** of a conveyance device **200** according to a second embodiment is described with reference to FIG. **9**. In the second embodiment, the magnetic scale **204** includes a magnetic shielding plate **243** to block the magnetism of each of linear movement magnetic scales **41** and curvilinear movement magnetic scales **42**, unlike the first embodiment. In the second embodiment, detailed description of the same or similar structures as those of the first embodiment is omitted.

(68) Base

(69) As shown in FIG. **9**, the magnetic scale **204** according to the second embodiment includes the magnetic shielding plate **243** located between the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** to block magnetism emitted from each of the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42**. The magnetic shielding plate **243** is made of ferromagnetic metal such as iron. Although the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are shown by broken lines in FIG. **9**, this is merely an image for visualizing the linear movement magnetic scales **41** and the

curvilinear movement magnetic scales **42** in the figure.

(70) The magnetic shielding plate **243** has a rectangular shape, the longitudinal direction of which is in an M direction as viewed from the Z1 direction side. The magnetic shielding plate **243** has a rectangular parallelepiped shape. The thickness of the magnetic shielding plate **243** in a Z direction is smaller than the width of the magnetic shielding plate **243** in a W direction. A base **240** on which a plurality of (three) linear movement magnetic scales **41** are recorded is attached to a first surface **243a** of the magnetic shielding plate **243**. The first surface **243a** is the upper surface of the magnetic shielding plate **243** on the Z1 direction side. A base **240** on which a plurality of (three) curvilinear movement magnetic scales **42** are recorded is attached to a second surface **243b** of the magnetic shielding plate **243**. The second surface **243b** is the lower surface of the magnetic shielding plate **243** on the Z2 direction side. Two, or four or more linear movement magnetic scales **41** may be recorded on the base **240**. Furthermore, two, or four or more curvilinear movement magnetic scales **42** may be recorded on the base **240**. The remaining structures of the second embodiment are the same as or similar to those of the first embodiment, and thus description thereof is omitted.

(71) According to the second embodiment, the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are spaced apart from each other, similarly to the first embodiment. Accordingly, magnetic interference between the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** can be more reliably reduced or prevented.

(72) According to the second embodiment, as described above, the magnetic scale **204** includes the magnetic shielding plate **243** located between the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** to block magnetism emitted from each of the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42**. Accordingly, magnetic interference between the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** can be reduced or prevented by the magnetic shielding plate **243**, and thus a distance by which the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are spaced apart from each other can be reduced. Consequently, thicknesses required for the bases **240** arranged between the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** can be reduced, and thus an increase in the size of the magnetic scale **204** including the linear movement magnetic scales **41**, the curvilinear movement magnetic scales **42**, and the bases **240** can be reduced or prevented. The remaining advantageous effects of the second embodiment are similar to the advantageous effects of the first embodiment.

Third Embodiment

(73) The structure of a magnetic scale **304** of a conveyance device **300** according to a third embodiment is described with reference to FIG. **10**. In the third embodiment, both linear movement magnetic scales **41** and curvilinear movement magnetic scales **42** are provided on the same surface of a base **340**, unlike the first embodiment. In the third embodiment, description of the same or similar structures as those of the first embodiment is omitted.

(74) Base

(75) As shown in FIG. **10**, the base **340** is a plate-shaped member made of a magnetic material using a ferrite material, for example. The base **340** has a rectangular shape, the longitudinal direction of which is in an M direction as viewed from the Z1 direction side. The base **340** has a rectangular parallelepiped shape. The thickness of the base **340** in a Z direction is smaller than the width of the base **340** in a W direction. Although the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are shown by broken lines in FIG. **10**, this is merely an image for visualizing the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** in the figure.

(76) Both the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are provided side by side on the same surface **340a** of the base **340**. The same surface **340a** is

the upper surface of the base **340**. The linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are provided on the same surface **340a** of the base **340** and are spaced apart from each other in a direction in which the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are aligned. Specifically, an isolation space **S** is provided between a group of a plurality of linear movement magnetic scales **41** and a group of a plurality of curvilinear movement magnetic scales **42**.

(77) According to the third embodiment, the following advantageous effects are achieved.

(78) According to the third embodiment, the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are spaced apart from each other, similarly to the first embodiment. Accordingly, magnetic interference between the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** can be more reliably reduced or prevented.

(79) According to the third embodiment, as described above, the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are provided on the same surface **340a** and are spaced apart from each other. Accordingly, the linear movement magnetic scales **41** and the curvilinear movement magnetic scale **42s** can be provided on the same surface **340a** of the same member, and thus as compared with a case in which the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are provided on surfaces of different members, an increase in the number of components of the conveyance device **300** can be reduced or prevented.

(80) According to the third embodiment, as described above, in the conveyance device **100**, the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are provided on the same surface **340a** of the base **340** and are spaced apart from each other in the direction in which the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are aligned. Accordingly, the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are provided on the same surface **340a** of the base **340** such that as compared with a case in which the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are aligned in a direction perpendicular to the same surface **340a**, an increase in a space for installing the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** in the direction perpendicular to the same surface **340a** can be reduced or prevented. The remaining advantageous effects of the third embodiment are similar to the advantageous effects of the first embodiment.

Fourth Embodiment

(81) The structure of a magnetic scale **404** of a conveyance device **400** according to a fourth embodiment is described with reference to FIG. **11**. In the fourth embodiment, a first surface **440a** and a second surface **440b** are substantially orthogonal to each other in a base **440**, unlike the first embodiment. In the fourth embodiment, description of the same or similar structures as those of the first embodiment is omitted.

(82) Base

(83) As shown in FIG. **11**, the base **440** is made of a magnetic material using a ferrite material, for example. The base **440** is an angle member having a substantially L-shape in a cross-section along a **Z** direction. Although linear movement magnetic scales **41** and curvilinear movement magnetic scales **42** are shown by broken lines in FIG. **11**, this is merely an image for visualizing the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** in the figure.

(84) The linear movement magnetic scales **41** are recorded on the first surface **440a** of the base **440**. The first surface **440a** is the inner side surface of the base **440**. The curvilinear movement magnetic scales **42** are recorded on the second surface **440b** of the base **440**. The second surface **440b** is the inner bottom surface of the base **440**. The second surface **440b** and the first surface **440a** are substantially orthogonal to each other.

(85) According to the fourth embodiment, the following advantageous effects are achieved.

(86) According to the fourth embodiment, the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are spaced apart from each other, similarly to the first

embodiment. Accordingly, magnetic interference between the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** can be more reliably reduced or prevented.

(87) According to the fourth embodiment, as described above, the second surface **440b** and the first surface **440a** are substantially orthogonal to each other. Accordingly, the linear movement magnetic scales **41** are provided on the first surface **440a**, and the curvilinear movement magnetic scales **42** are provided on the second surface **440b** that is substantially orthogonal to the first surface **440a**, and thus magnetic sensors **5** that detect the magnetism of the linear movement magnetic scales **41** can be placed in the vicinity of magnetic sensors **6** that detect the magnetism of the curvilinear movement magnetic scales **42**. Consequently, a space required to place the magnetic sensors **5** and the magnetic sensors **6** can be made compact. The remaining advantageous effects of the fourth embodiment are similar to the advantageous effects of the first embodiment.

Fifth Embodiment

(88) The structure of a magnetic scale **504** of a conveyance device **500** according to a fifth embodiment is described with reference to FIG. **12**. In the fifth embodiment, a second surface **540b** and a first surface **540a** are substantially orthogonal to each other, unlike the first embodiment. In the fifth embodiment, description of the same or similar structures as those of the first embodiment is omitted.

(89) Base

(90) As shown in FIG. **12**, a base **540** is made of a magnetic material using a ferrite material, for example. Although linear movement magnetic scales **41** and curvilinear movement magnetic scales **42** are shown by broken lines in FIG. **12**, this is merely an image for visualizing the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** in the figure.

(91) The linear movement magnetic scales **41** are recorded on the first surface **540a** of the base **540**. The first surface **540a** is the upper surface of the base **540**. The curvilinear movement magnetic scales **42** are recorded on the second surface **540b** of the base **540**. The second surface **540b** is the outer side surface of the base **540**. The second surface **540b** and the first surface **540a** are substantially orthogonal to each other. The second surface **540b** is a curved surface corresponding to the curvature of the curvilinear movement magnetic scales **42**.

(92) According to the fifth embodiment, the following advantageous effects are achieved.

(93) According to the fifth embodiment, the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are spaced apart from each other, similarly to the first embodiment. Accordingly, magnetic interference between the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** can be more reliably reduced or prevented.

(94) According to the fifth embodiment, as described above, the second surface **540b** and the first surface **540a** are substantially orthogonal to each other. The second surface **540b** is a curved surface corresponding to the curvature of the curvilinear movement magnetic scales **42**.

Accordingly, the curved surface of the second surface **540b** is formed along the curvilinear direction of a slider **3**, and thus the position of the slider **3** during curvilinear movement can be appropriately detected by the curvilinear movement magnetic scales **42** provided on the second surface **540b**. The remaining advantageous effects of the fifth embodiment are similar to the advantageous effects of the first embodiment.

Modified Examples

(95) The embodiments disclosed this time must be considered as illustrative in all points and not restrictive. The scope of the present disclosure is not shown by the above description of the embodiments but by the scope of claims for patent, and all modifications (modified examples) within the meaning and scope equivalent to the scope of claims for patent are further included.

(96) For example, while the example in which in the magnetic scale **4** (**204**, **304**, **404**, and **504**), the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are provided integrally with the base **40** (**240**, **340**, **440**, and **540**) has been shown in each of the aforementioned first to fifth embodiments, the present disclosure is not restricted to this. In the present disclosure,

as in a first modified example shown in FIG. 13, one set of linear movement magnetic scales **41** and two sets of curvilinear movement magnetic scales **42** may be provided integrally not with a base **40** but with a base member **640**. Specifically, a conveyance device **600** includes a magnetic scale **604** and a back yoke **607**. The magnetic scale **604** includes the base member **640** in which linear movement magnetic scales **41** and curvilinear movement magnetic scales **42** are provided. The curvilinear movement magnetic scales **42** include first curvilinear scales **642a** formed along a first curvilinear direction as a curvilinear direction, and second curvilinear scales **642b** formed along a second curvilinear direction different from the first curvilinear direction. The base member **640** includes a linear position detection surface **640a** that constitutes the lower surface of the base member **640** and on which the linear movement magnetic scales **41** are provided. The base member **640** includes a first curvilinear position detection surface **640b** that protrudes from the linear position detection surface **640a** and on which the first curvilinear scales **642a** are provided. The base member **640** includes a second curvilinear position detection surface **640c** that faces the first curvilinear position detection surface **640b** and on which the second curvilinear scales **642b** are provided. Accordingly, with the first curvilinear scales **642a** and the second curvilinear scales **642b**, the position of a slider **3** can be measured when the slider **3** moves curvilinearly in a plurality of curvilinear directions including the first curvilinear direction and the second curvilinear direction, and thus the position measurement can be performed even when the slider **3** moves curvilinearly in curvilinear directions that are opposite to each other. Thus, the degree of freedom of a path along which the slider **3** moves can be improved. The magnetic scale **604** may be formed by magnetizing each of the linear movement magnetic scales **41**, the first curvilinear scales **642a**, and the second curvilinear scales **642b** to a magnetic material and then bonding it to the base member **640**. Alternatively, the magnetic scale **604** may be formed by magnetizing each of the linear movement magnetic scales **41**, the first curvilinear scales **642a**, and the second curvilinear scales **642b** to a magnetic body bonded to the base member **640**. Alternatively, the magnetic scale **604** may be formed by insert-molding a plurality of magnetic materials to which the linear movement magnetic scales **41**, the first curvilinear scales **642a**, and the second curvilinear scales **642b** have been magnetized into the base member **640**.

(97) While the example in which the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are provided integrally with the base **40** (**240**, **340**, **440**, and **540**) has been shown in each of the aforementioned first to fifth embodiments, the present disclosure is not restricted to this. In the present disclosure, the magnetic material to which the linear movement magnetic scales and the curvilinear movement magnetic scales have been magnetized may be bonded to the base.

(98) While the example in which the magnetic scale **4** (**204**, **304**, **404**, and **504**) is provided on the slider **3** has been shown in each of the aforementioned first to fifth embodiments, the present disclosure is not restricted to this. In the present disclosure, the magnetic scale may be provided on the drive module. In this case, the magnetic sensor is provided in the slider instead of the drive module.

(99) While the example in which the magnetic scale **4** (**204**, **304**, **404**, and **504**) is arranged inside the slider frame **31** has been shown in the aforementioned first embodiment, the present disclosure is not restricted to this. In the present disclosure, the magnetic scale may be arranged outside the slider frame.

(100) While the example in which the base **40** (**240**, **340**, **440**, and **540**) is a thin plate, the thickness of which is smaller than its width in the Y direction, has been shown in each of the aforementioned first to third embodiments, the present disclosure is not restricted to this. In the present disclosure, the base may be a thin plate, the thickness in the Y direction of which is smaller than its width in the Z direction.

(101) While the example in which the same surface **340a** is the upper surface of the base **40** (**240**, **340**, **440**, and **540**) has been shown in the aforementioned third embodiment, the present disclosure

is not restricted to this. In the present disclosure, the same surface may be a lower surface or a side surface, for example, other than the upper surface.

(102) While the example in which the conveyance device **100** (**200**, **300**, **400**, **500**) is a linear conveyor that moves the slider **3** along the circular path **102** using a linear motor has been shown in each of the aforementioned first to fifth embodiments, the present disclosure is not restricted to this. In the present disclosure, the conveyance device may drive the slider by driving a belt conveyor. Furthermore, the conveyance device may be a device combined with a conveyance mechanism that conveys the entire slider in a direction orthogonal to the linear movement direction of the slider.

(103) While the example in which the path along which the slider **3** moves is an annular circular path **102** provided in a horizontal plane has been shown in each of the aforementioned first to fifth embodiments, the present disclosure is not restricted to this. In the present disclosure, the path along which the slider moves may have a slope inclined with respect to the horizontal plane.

(104) While the example in which the path along which the slider **3** moves is a circular path **102** that is circular in the plan view has been shown in each of the aforementioned first to fifth embodiments, the present disclosure is not restricted to this. In the present disclosure, the path along which the slider moves may not be a circular path, but may be a J-shaped path in the plan view.

(105) While the example in which three linear movement magnetic scales **41** and three curvilinear movement magnetic scales **42** are provided has been shown in each of the aforementioned first to fifth embodiments, the present disclosure is not restricted to this. In the present disclosure, two or four or more linear movement magnetic scales and two or four or more curvilinear movement magnetic scales may be provided.

(106) While the example in which the base **440** is an angle member having a substantially L-shape in the cross-section along the Z direction (upward-downward direction) has been shown in the aforementioned fourth embodiment, the present disclosure is not restricted to this. In the present disclosure, the base may be a substantially C-shaped member having a first surface and a second surface that face each other in the upward-downward direction and a connecting surface that connects the first surface to the second surface.

(107) While the example in which the second surface **440b** (**562**) and the first surface **440a** (**561**) are substantially orthogonal to each other has been shown in each of the aforementioned fourth and fifth embodiments, the present disclosure is not restricted to this. In the present disclosure, the second surface and the first surface may intersect with each other at an acute angle or an obtuse angle.

(108) While the example in which the magnetic scale **4** is arranged inside the slider frame **31** with the linear movement magnetic scales **41** facing the magnetic sensors **5** in the Z direction, and the curvilinear movement magnetic scales **42** facing the magnetic sensors **6** in the Z direction has been shown in the aforementioned first embodiment, the present disclosure is not restricted to this. The magnetic scale may be arranged inside the slider frame with the linear movement magnetic scales facing the magnetic sensors in the width direction, and the curvilinear movement magnetic scales facing the magnetic sensors in the width direction.

(109) While the example in which the base **240** on which the plurality of (three) linear movement magnetic scales **41** are recorded is attached to the first surface **243a** of the magnetic shielding plate **243**, and the base **240** on which the plurality of (three) curvilinear movement magnetic scales **42** are recorded is attached to the second surface **243b** of the magnetic shielding plate **243** has been shown in the aforementioned second embodiment, the present disclosure is not restricted to this. In the present disclosure, as in a second modified example shown in FIG. **14**, a plurality of (three) bases **740** on which linear movement magnetic scales **41** are recorded may be attached to a first surface **243a** of a magnetic shielding plate **243**. Furthermore, a plurality of (three) bases **740** on which curvilinear movement magnetic scales **42** are recorded may be attached to a second surface

243b of the magnetic shielding plate **243**. Alternatively, two, or four or more bases **740** on which the linear movement magnetic scales **41** are recorded may be attached to the first surface **243a** of the magnetic shielding plate **243**. Furthermore, two, or four or more bases **740** on which the curvilinear movement magnetic scales **42** are recorded may be attached to the second surface **243b** of the magnetic shielding plate **243**. Although the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are shown by broken lines in FIG. **14**, this is merely an image for visualizing the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** in the figure.

(110) While the example in which the base **240** on which the plurality of (three) linear movement magnetic scales **41** are recorded is attached to the first surface **243a** of the magnetic shielding plate **243**, and the base **240** on which the plurality of (three) curvilinear movement magnetic scales **42** are recorded is attached to the second surface **243b** of the magnetic shielding plate **243** has been shown in the aforementioned second embodiment, the present disclosure is not restricted to this. In the present disclosure, as in a third modified example shown in FIGS. **15** and **16**, a base **840** on which a plurality of (three) linear movement magnetic scales **41** and a plurality of (three) curvilinear movement magnetic scales **42** are recorded may be insert-molded into a magnetic shielding plate **243**. Alternatively, two, or four or more linear movement magnetic scales **41** may be recorded on the base **840**. Furthermore, two, or four or more curvilinear movement magnetic scales **42** may be recorded on the base **840**. Although the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** are shown by broken lines in FIGS. **15** and **16**, this is merely an image for visualizing the linear movement magnetic scales **41** and the curvilinear movement magnetic scales **42** in the figures.

Claims

1. A conveyance device comprising: a slider configured to move along a linear direction and a curvilinear direction; a drive module configured to move the slider along the linear direction or the curvilinear direction; a magnetic scale including linear movement magnetic scales configured linearly along the linear direction, and curvilinear movement magnetic scales configured curvilinearly along the curvilinear direction and provided on either the slider or the drive module; and magnetic sensors in either the drive module or the slider and configured to detect magnetism of the linear movement magnetic scales and the curvilinear movement magnetic scales, respectively; wherein the linear movement magnetic scales and the curvilinear movement magnetic scales are spaced apart from each other.
2. The conveyance device according to claim 1, wherein the linear movement magnetic scales and the curvilinear movement magnetic scales are on a same surface or different surfaces and are spaced apart from each other.
3. The conveyance device according to claim 1, wherein the magnetic scale is on the slider; and the magnetic sensors are in the drive module.
4. The conveyance device according to claim 3, wherein the slider includes a slider frame in which the magnetic scale is located such that the linear movement magnetic scales and the curvilinear movement magnetic scales face the magnetic sensors in an upward-downward direction or in a width direction orthogonal to the upward-downward direction and a moving direction of the slider.
5. The conveyance device according to claim 1, wherein the magnetic scale further includes a base with which the linear movement magnetic scales and the curvilinear movement magnetic scales are integral.
6. The conveyance device according to claim 5, wherein the base includes: a first surface on which the linear movement magnetic scales are provided; and a second surface that is different from the first surface and on which the curvilinear movement magnetic scales are provided.
7. The conveyance device according to claim 6, wherein the second surface is on a back side of the

first surface in the base.

8. The conveyance device according to claim 7, wherein the base includes a magnetic shielding plate between the linear movement magnetic scales and the curvilinear movement magnetic scales to block magnetism emitted from each of the linear movement magnetic scales and the curvilinear movement magnetic scales.

9. The conveyance device according to claim 6, wherein the second surface and the first surface are substantially orthogonal to each other.

10. The conveyance device according to claim 9, wherein the second surface and the first surface are substantially orthogonal to each other; and the second surface is a curved surface corresponding to a curvature of the curvilinear movement magnetic scales.

11. The conveyance device according to claim 5, wherein the linear movement magnetic scales and the curvilinear movement magnetic scales are on a same surface of the base and are spaced apart from each other in a direction in which the linear movement magnetic scales and the curvilinear movement magnetic scales are aligned.

12. The conveyance device according to claim 1, wherein the magnetic scale further includes a base member in which the linear movement magnetic scales and the curvilinear movement magnetic scales are provided; the curvilinear movement magnetic scales include: first curvilinear scales configured along a first curvilinear direction as the curvilinear direction; and second curvilinear scales configured along a second curvilinear direction different from the first curvilinear direction; the base member includes: a linear position detection surface that configures an upper surface or a lower surface of the base member and on which the linear movement magnetic scales are provided; a first curvilinear position detection surface that protrudes from the linear position detection surface and on which the first curvilinear scales are provided; and a second curvilinear position detection surface that faces the first curvilinear position detection surface and on which the second curvilinear scales are provided.

13. The conveyance device according to claim 1, wherein the magnetic scale is on the slider; and in a moving direction of the slider, first ends of the linear movement magnetic scales and first ends of the curvilinear movement magnetic scales are at substantially a same position, and second ends of the linear movement magnetic scales and second ends of the curvilinear movement magnetic scales are at substantially a same position.

14. The conveyance device according to claim 2, wherein the magnetic scale is on the slider; and the magnetic sensors are in the drive module.

15. The conveyance device according to claim 2, wherein the magnetic scale further includes a base with which the linear movement magnetic scales and the curvilinear movement magnetic scales are integral.

16. The conveyance device according to claim 3, wherein the magnetic scale further includes a base with which the linear movement magnetic scales and the curvilinear movement magnetic scales are integral.

17. The conveyance device according to claim 4, wherein the magnetic scale further includes a base with which the linear movement magnetic scales and the curvilinear movement magnetic scales are integral.

18. The conveyance device according to claim 2, wherein the magnetic scale further includes a base member in which the linear movement magnetic scales and the curvilinear movement magnetic scales are provided; the curvilinear movement magnetic scales include: first curvilinear scales configured along a first curvilinear direction as the curvilinear direction; and second curvilinear scales configured along a second curvilinear direction different from the first curvilinear direction; the base member includes: a linear position detection surface that configures an upper surface or a lower surface of the base member and on which the linear movement magnetic scales are provided; a first curvilinear position detection surface that protrudes from the linear position detection surface and on which the first curvilinear scales are provided; and a second curvilinear

position detection surface that faces the first curvilinear position detection surface and on which the second curvilinear scales are provided.

19. The conveyance device according to claim 3, wherein the magnetic scale further includes a base member in which the linear movement magnetic scales and the curvilinear movement magnetic scales are provided; the curvilinear movement magnetic scales include: first curvilinear scales configured along a first curvilinear direction as the curvilinear direction; and second curvilinear scales configured along a second curvilinear direction different from the first curvilinear direction; the base member includes: a linear position detection surface that configures an upper surface or a lower surface of the base member and on which the linear movement magnetic scales are provided; a first curvilinear position detection surface that protrudes from the linear position detection surface and on which the first curvilinear scales are provided; and a second curvilinear position detection surface that faces the first curvilinear position detection surface and on which the second curvilinear scales are provided.

20. The conveyance device according to claim 2, wherein the magnetic scale is on the slider; and in a moving direction of the slider, first ends of the linear movement magnetic scales and first ends of the curvilinear movement magnetic scales are at substantially a same position, and second ends of the linear movement magnetic scales and second ends of the curvilinear movement magnetic scales are at substantially a same position.
